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Explaining determinants of bank failure prediction via neural additive model

Bumho Son^a, Jaewook Lee^b and Hoki Kim^c

^aDepartment of Business Administration, College of Business & Economics, Chung-Ang University, Seoul, South Korea; ^bDepartment of Industrial Engineering, Seoul National University, Seoul, South Korea; ^cDepartment of Industrial Security, College of Business & Economics, Chung-Ang University, Seoul, South Korea

ABSTRACT

This study aims to explain the determinants of bank failures by evaluating existing prediction models. We reveal that existing machine learning models often result in inconsistent explanations within the financial domain, leading to flawed conclusions and suboptimal decision-making. Therefore, we propose the use of a neural additive model that combines the transparency of a general additive model with the high performance of a neural network. Our findings show that this approach enhances the interpretation of explanatory variables in bank failure predictions, offering valuable insights for improving financial risk assessment.

KEYWORDS

Bank failure prediction; machine learning approaches; financial risk indicator; neural additive model

JEL CLASSIFICATION

G21; G01

I. Introduction

Bankruptcies, despite being infrequent, have substantial ramifications in the financial sector, making their prediction crucial (Looney, Wansley, and Lane 1989; Meyer and Pifer 1970). In recent years, advancements in machine learning have improved the performance of bankruptcy prediction using large volumes of historical data. However, their black-box nature remains a serious problem in finance, where transparency and accountability are vital for regulatory compliance. While several post-hoc techniques are often used to uncover the determinants of predictions in the finance domain (Babaei et al. 2022; Lin and Bai 2022), recent studies (Kumar et al. 2020; Rudin 2019) have demonstrated that they can produce misleading explanations within the vision domain.

In this article, we build on findings from previous studies (Kumar et al. 2020; Rudin 2019) to examine the risks associated with using black-box models and post-hoc methods for explaining determinants of bank failures. We then propose a neural additive model that provides interpretability alongside strong predictive performance. This study contributes to academic discourse by exposing the limitations of post-hoc methods in financial prediction tasks and by demonstrating a potential approach to balance model accuracy and

transparency, thus laying a foundation for more accountable machine learning practices in finance.

This article is organized as follows: [Section II](#) provides an overview of the methodologies and background. [Section IV](#) describes the data used for the empirical analysis. In [Section V](#), we present a summary of the empirical results, focusing on the pitfalls of post-hoc methods regarding inconsistent determinants and explaining the determinants of bank failure prediction. Finally, [Section 5](#) concludes our findings.

II. Data and method

Data

We focused on the bank data listed on the New York Stock Exchange (NYSE), American Stock Exchange (AMEX), and National Association of Securities Dealers Automated Quotations (NASDAQ). The data range was from January 1969 to September 2021, and the total dataset consisted of 21,243 banks. As explanatory variables, Meyer and Pifer (1970), initially proposed internal characteristics of banks, such as return on equity (ROE), return on asset (ROA), and the ratio of operating revenue to operating cost. Similar variables – capital, assets, and liquidity – and several external conditions have been

Table 1. Summary statistics of the dataset.

Type	Name	Mean	St. Dev.	Min	Q1	Q3	Max
Solvency Ratio	Leverage Ratio	11.0	4.49	0.45	8.13	13.13	101.37
	Asset Liabilities	1.11	0.08	1.02	1.08	1.12	3.28
	Debt Ratio	0.90	0.04	0.31	0.89	0.93	0.98
	Capitalization Ratio	0.38	0.22	0.00	0.19	0.55	0.95
	Long-term Debt/Invested Capital	0.39	0.26	0.00	0.20	0.56	3.05
	Total Debt/Invested Capital	0.86	0.67	0.01	0.45	1.04	5.41
	Debt/Ebitda	6.82	14.95	-207.98	2.46	8.99	402.72
Profitability Ratio	ROE	0.09	0.16	-2.38	0.06	0.14	3.53
	ROA	0.02	0.01	-0.09	0.02	0.03	0.21
Efficiency Ratio	Asset Turnover	0.08	0.04	0.02	0.05	0.09	0.92
	Receivables Turnover	0.15	0.47	0.04	0.08	0.15	17.77
Coverage Ratio	Cash/Debt	0.01	0.05	-0.54	0.00	0.02	0.38

widely used in prior research (Alzayed, Eskandari, and Yazdifar 2023; Cebula 2010; Shrivastava et al. 2020). Building on prior research, we carefully selected 12 widely used and significant explanatory variables across categories such as solvency, profitability, efficiency, and coverage ratios. The raw data were collected from COMPUSTAT. For the out-of-sample analysis, we used 80% of the total data as the training set and the rest as the test set. Table 1 provides a summary of the explanatory variables.

Method

The basic linear model for bankruptcy prediction can be expressed as follows:

$$Prob(y_n = 1|x_{n,k}) = \sum_{k=1}^K \beta_k x_{n,k} + \alpha, \quad (1)$$

where $x_{n,k}$ is a set of explanatory variables for $k \in \{1, \dots, K\}$, and $Prob(y_n = 1|x_{n,k})$ is the probability of bankruptcy for each observation n . In the field of bank failure prediction, the goal is to accurately predict the target variable y_n , which represents the failure of a bank, i.e. 'Failed' or 'Non-failed' (Meyer and Pifer 1970).

In the mid-1970s, most models for bank failure prediction were constructed using classical statistical techniques (Looney, Wansley, and Lane 1989), later expanding to simple and interpretable machine learning methods, such as logistic regression (Martin 1977) and decision trees. While more complex models such as Support Vector Machines (SVM) and Multi-Layer Perceptrons (MLP) (Abakarim, Lahby, and Attiou 2020) achieve high performance, their limited interpretability presents a significant challenge. Therefore, post-hoc methods, which are additionally used for the trained

model, have been adopted to find the determinants of predictions in the finance domain (Babaei et al. 2022; Lin and Bai 2022).

However, recent studies in the vision domain have argued that these post-hoc methods often result in unreliable explanations (Rudin 2019), suggesting instead the use of intrinsically interpretable models. Among them, the neural additive model (NAM) (Agarwal et al. 2021) has gained attention for its intrinsic interpretability. For the task of bank failure prediction (1), it can be formalized as follows:

$$\begin{aligned} y_n &= f_1(x_{n,1}) + f_2(x_{n,2}) + \dots + f_K(x_{n,K}) + \alpha \\ &= \sum_{k=1}^K f_k(x_{n,k}) + \alpha. \end{aligned} \quad (2)$$

Here, each function f_k is constructed with a neural network. Prior work (Jo and Kim 2023; Kraus et al. 2024) confirms that NAM also achieves both high predictive accuracy and explainability in various domains including time series. Therefore, we posit that the structural simplicity of NAM can also provide explainable benefits similar to linear regression while further improving bank failure prediction.

In our study, we use logistic regression, decision tree, SVM with linear kernel and radial basis function, XGBoost, and MLP. Each method undergoes hyper-parameter tuning using grid search. For our approach, we employ a neural network structure with layers of (64, 64, 32). Table 2 presents the results for precision (Prec.), recall, F1 score, accuracy (Acc.), and area under the curve (AUC). Among explainable models, our model outperforms logistic regression and decision tree models across all performance measures. Notably, our

Table 2. Comparison of prediction performance of models.

Model	Explainable	TP	FP	FN	TN	Prec.	Recall	F1	Acc.	AUC
Logistic Reg.	YES	4359	887	23	96	0.10	0.81	0.17	0.83	0.89
Decision Tree	YES	4090	1156	25	94	0.08	0.79	0.14	0.78	0.91
Ours	YES	4429	817	17	102	0.11	0.86	0.20	0.84	0.93
SVM-Linear	NO	3631	1615	11	108	0.06	0.91	0.12	0.70	0.91
SVM-RBF	NO	4063	1183	15	104	0.08	0.87	0.15	0.78	0.92
XGBoost	NO	4745	501	25	94	0.16	0.79	0.26	0.90	0.92
MLP	NO	4775	471	27	92	0.16	0.77	0.27	0.91	0.93

The best results are in bold for evaluation metrics.

model also shows comparable performance to other non-explainable models such as SVM and MLP.

III. Results

Figure 1 demonstrates the difficulty of using existing machine learning methods in the finance domain. For SVM and MLP, we applied the commonly used explanatory tools in the finance domain, SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME). For MLP, we also adapt

Integrated Gradient (IG). To ease the comparison, we normalize the importance of each feature to a range between -1 and 1 . In the top figure, SHAP and LIME offer inconsistent explanations, with SHAP identifying ‘Cash/Debt’ as the most influential feature, while LIME points to ‘Asset Liabilities’. Similar inconsistencies are observed with MLP. These inconsistent explanations highlight the potential danger of using post-hoc methods for black-box models in the finance domain.

Figure 2 demonstrates the contributions of our model for failed and non-failed banks. Following Agarwal et al. (2021), we calculate the contribution

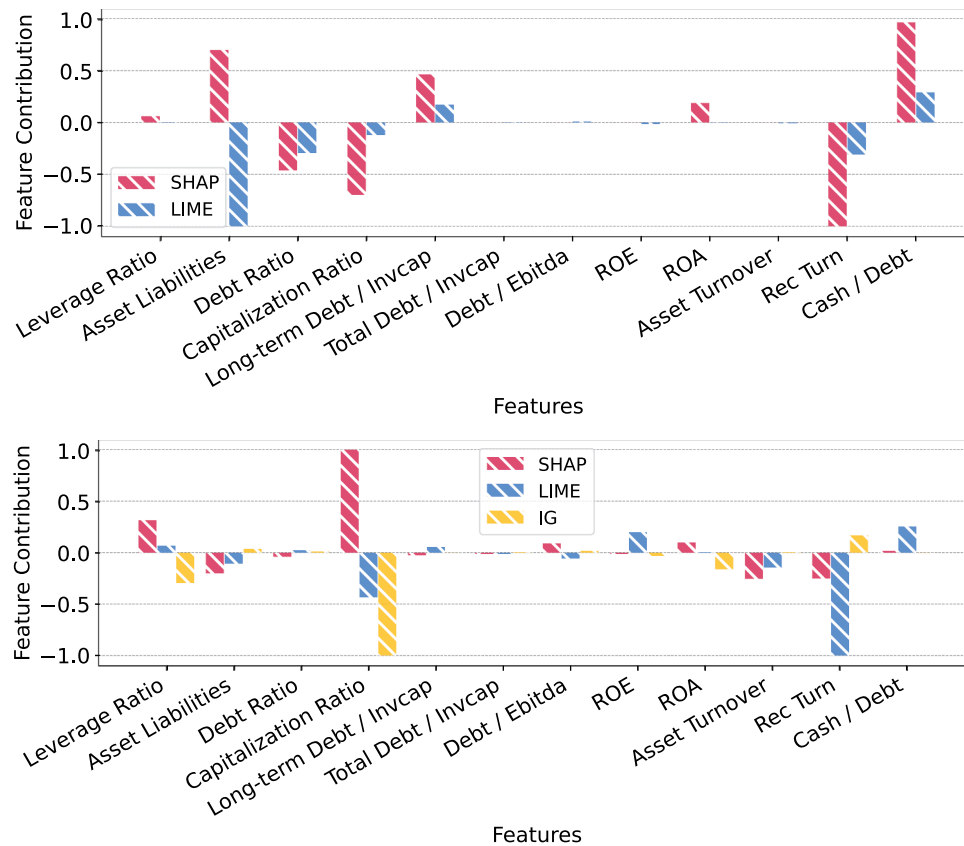


Figure 1. Contribution of each explanatory variable with various post-hoc methods. A positive contribution of a variable implies that it increases the likelihood of failure. Despite the same SVM (top) and MLP (bottom) being used, the explanations generated by these methods are significantly different.

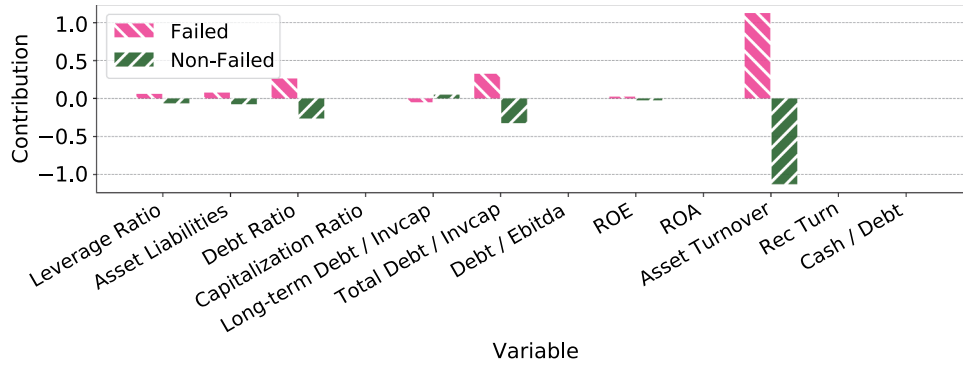


Figure 2. Contribution of each explanatory variable to the target variable of ‘failed’ ($y = 1$) and ‘non-failed’ ($y = 0$). A positive contribution of a variable indicates that the variable increases the probability of bank failure. Our method provides a consistent explanation.

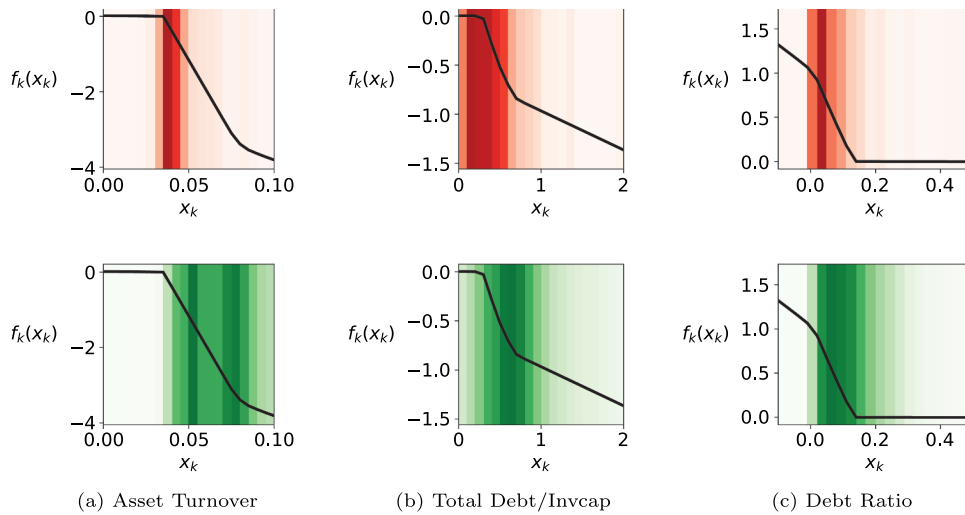


Figure 3. Function shapes $f_k(x_k)$ of important explanatory variables x_k and distributions of x_k in high-density regions. The background colour corresponds to the distributions of failed banks (top) and non-failed banks (bottom), with darker colours indicating higher density.

by $f_k(x_k) - \sum_{n=1}^N [f_k(x_{n,k})]/N$ of each feature x_k for all N samples. A positive contribution of a variable suggests it increases the likelihood of bank failure, while a negative contribution indicates it reduces this likelihood. ‘Asset turnover’, ‘Total-debt/Invcap’, and ‘Debt Ratio’ have high contributions for both failed banks and non-failed banks. This result is well aligned with previous findings that cash flow and debt ratios are significant variables for bankruptcy prediction (Beaver 1966).

Figure 3 illustrates the function shapes of the variables identified in Figure 1. Figure 2 shows that ‘Asset Turnover’ decreases as its value increases, indicating that lower asset turnover correlates with a higher probability of bank failure, consistent with

(Manousaridis 2017). The background distributions further support this, with failed banks clustered on the left side of 0.05. Similar downward-sloping patterns are observed for ‘Total Debt/Invested Capital’ and ‘Debt Ratio’, where lower values increase the likelihood of bank failure, aligning with (Zaghdoudi 2013) that highlighted the role of debt, including deposits, in bank stability.

To validate the importance of determinants identified by our model, we removed the top three variables from each model and then measured their AUCs across 100 random splits with logistic regression models. As shown in Table 3, our model exhibited the largest AUC decrease, indicating that it more accurately

Table 3. Effect of eliminating the top-3 important variables selected by explainable models.

	$X - X_{LR}$	$X - X_{DT}$	$X - X_{Ours}$
Panel A: Reduction of AUC when important variables selected by each model are eliminated.			
ΔAUC	0.130 (−149.319**)	−0.009 (8.957***)	0.163 (−205.041***)
Panel B: Pair-wise t-test for the results comparison in Panel A.			
$X - X_{LR}$	0.000	−0.139 (120.965***)	0.033 (−35.163***)
$X - X_{DT}$	−	0.000	0.172 (−200.397***)
$X - X_{Ours}$	−	−	0.000

In Panel A, we measure the reduction of AUC (ΔAUC) after the elimination of top-3 important variables selected by each model. A larger reduction indicates that the eliminated variables are more important. In Panel B, for each cell, a larger value implies that the reduction in AUC, resulting from the elimination of variables selected by the model in the column, is greater than that in the row. For both panels, we trained 100 logistic regression models using the given variables. The paired t-test statistic is reported in parentheses, and *** indicates that the p-value is smaller than 0.01.

identifies key determinants compared to other models.

IV. Conclusion

Our study reveals the limitations of post-hoc explanations in black-box models for bank failure prediction. Through empirical analysis, we confirm that a neural additive model not only enhances predictive performance but also provides reliable key determinants of bank failure. Finally, we offer insights into the role of each predictor, linking them to traditional analyses in bank failure prediction.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The data used in this work are available from COMPUSTAT.

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